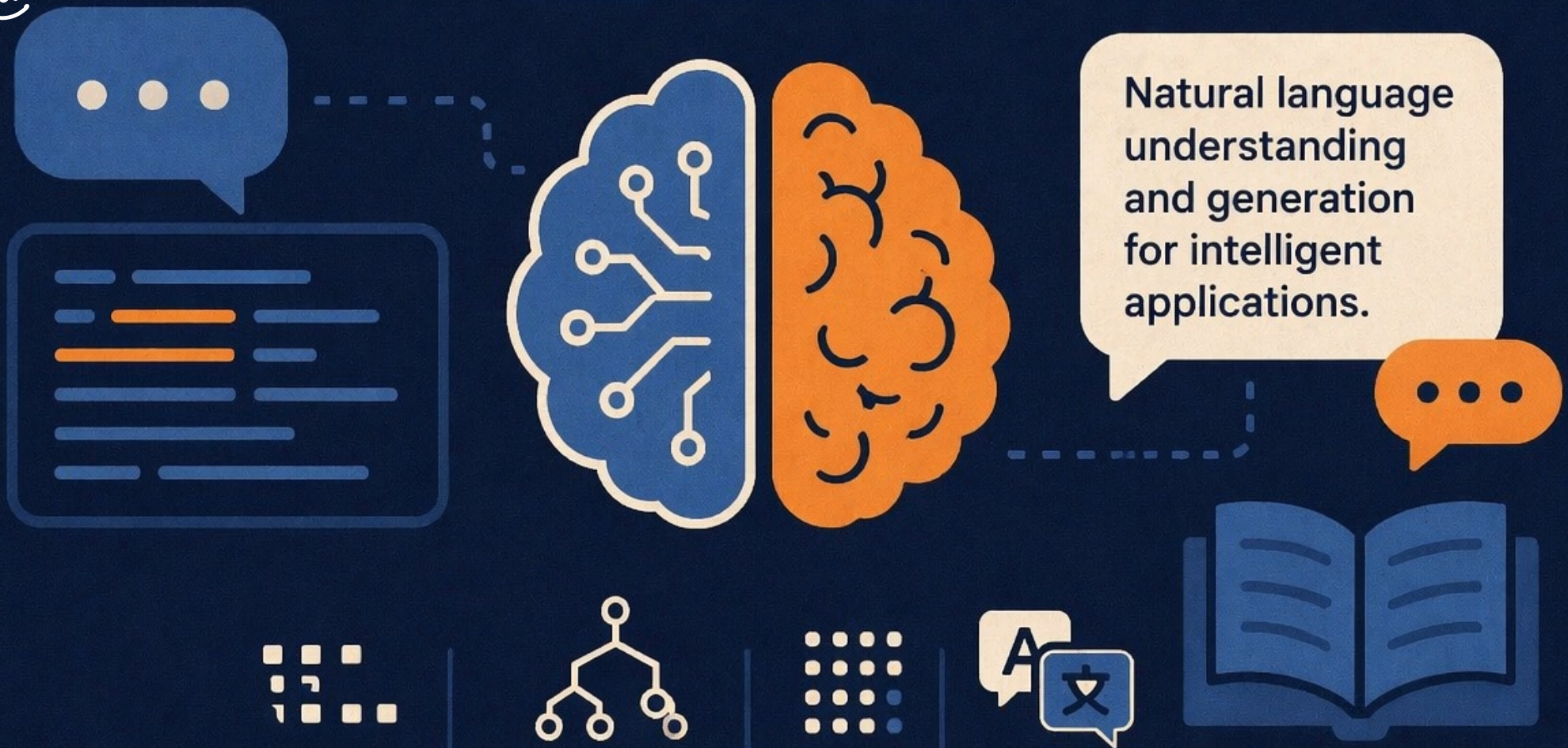
# NATURAL LANGUAGE PROCESSING

class starts @ 9:05 pm



# Lecture: Topic Modelling and RNN

#Agenda

1) Topic Modelling
2) RNN

# #Recap

1) Sentiment Analysis (Task)

↳ To capture sentiment of a sentence

a) TextBlob → polarity , subjectivity

polarity → +1/-1 (range) → pos / -neg

subjectivity → 0-1 → More subject / more factual

NLP

1) Corpus → Collection of docs
2) documents → Collection of sentences
3) Sentence
4) Word

b) NRC based

↳ Emotions

c) AFINN

↳ Scale of +5 to -5

for each word of a sentence & then collectively scoring it.

# What is Topic Modelling?

Documents → (reviews, pdf, ...)

↳ 1000s of review

What type of topic is being discussed upon?

# What is a Topic?

→ Theme of a document → {win, rally, candidate, majority}

{sports, politics, technology}

{ cricket, ball, win, score, football, net }

Topic: A cluster of related words

A document: a mix of multiple topics

# LDA (Latent Dirichlet Allocation)

↳ Name of a distribution

Bunch of documents (articles, reviews, tweets)

LDA assumes

1) Each documents is a mix of topics

2) Each topic is a mix of words

So instead of saying,

"This document is about sports"

LDA says,

"This document is 70% sports, 20% politics, 10% health"

[0.7 0.2 0.10]

Dirichlet ⇒ Total prob dist sums up to 1

# Example

Suppose we have 3 topics

1) Topic 1: sports → {game, team, score, player}
2) Topic 2: politics → {election, vote, government}
3) Topic 3: health → {doctor, medicine, hospital}

document → "The team won the game & the player scored"

LDA might says, 95% sports, 4% politics, 1% health

Combination of

probability distribution of words + Co-occurce of words

I won the game

won game

(conditional based)

Team A won the match by 3 runs

Team A won match 3 runs

P(won | game)  P(won | match)


Hello my name Rahul
P(Hello | my) ...

# How does LDA imagines documents were generated.

→ Its a generative model

For each document:

1) Pick a distribution of topics (eg: 70% sports, 30% politics)

2) For each word:

→ Choose a topic (based on the document mix)

→ Chose a word from that topic word distribution

LDA

↳ 2 dirichlet distribution

1) Document → Topic

2) Topic → Word

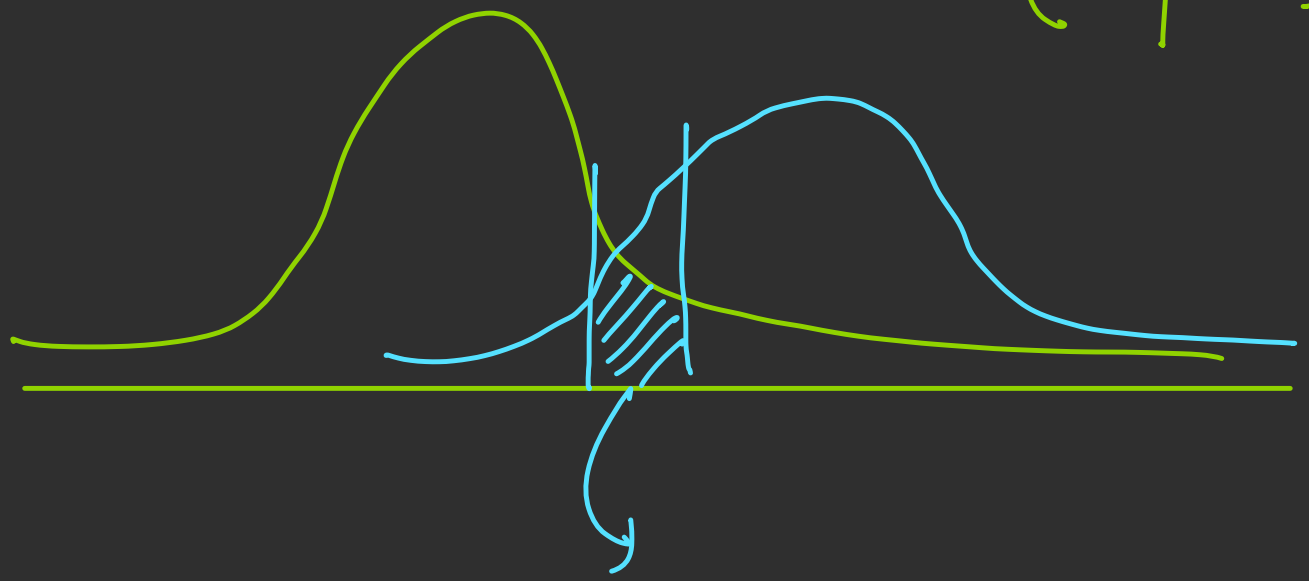
Document → Topic ← word

↳ [win, loss, ...]

↳ all unique word

⋮

P(win | loss) = w



# RNN

↳ Recurrent Neural Network

ANN — Tabular Data

CNN — grid structure image

RNN — Sequential Data

Timeseries data

DNA pattern

# Why we need RNN?

ANN

Text classification

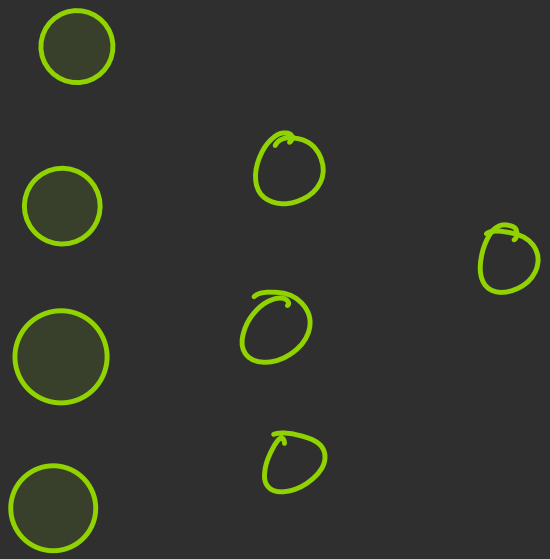
Email

king needs help

Target

Spam

Ham



Binary classi

# Why we need RNN?

1) Text length are not same.

2) The idea of sequence is lost.

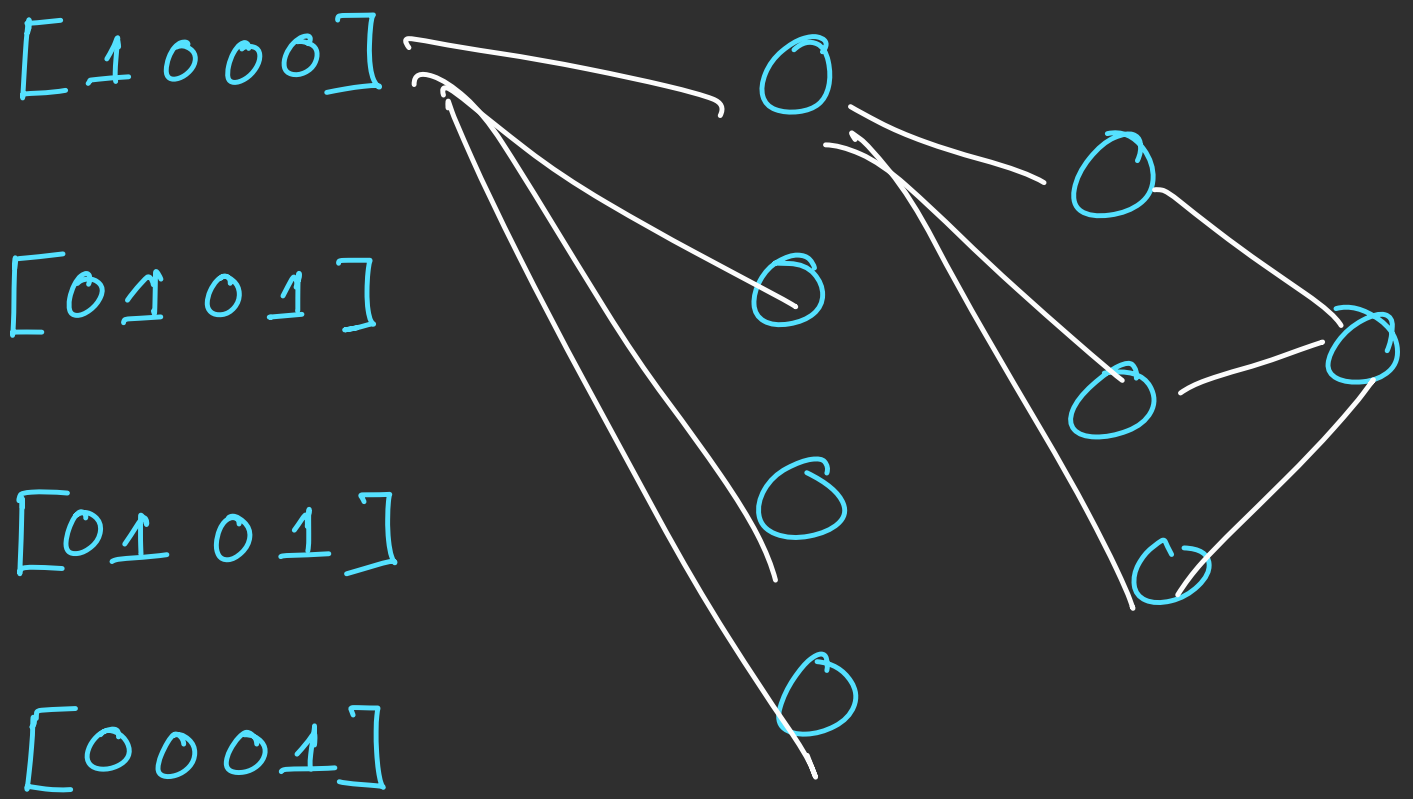
Need m..

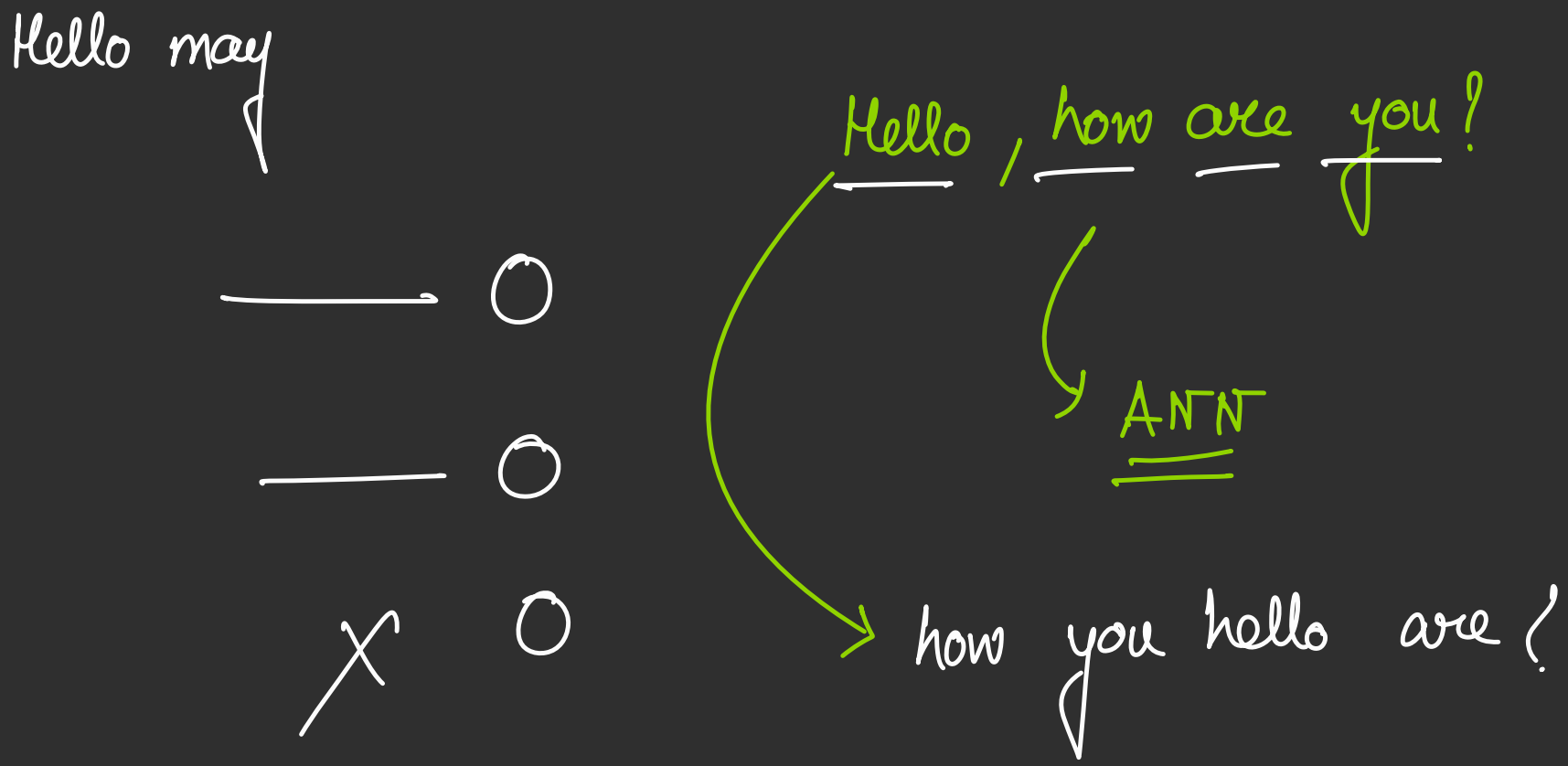
Text ⟶ vector rep → OHE [1 0 0 0]

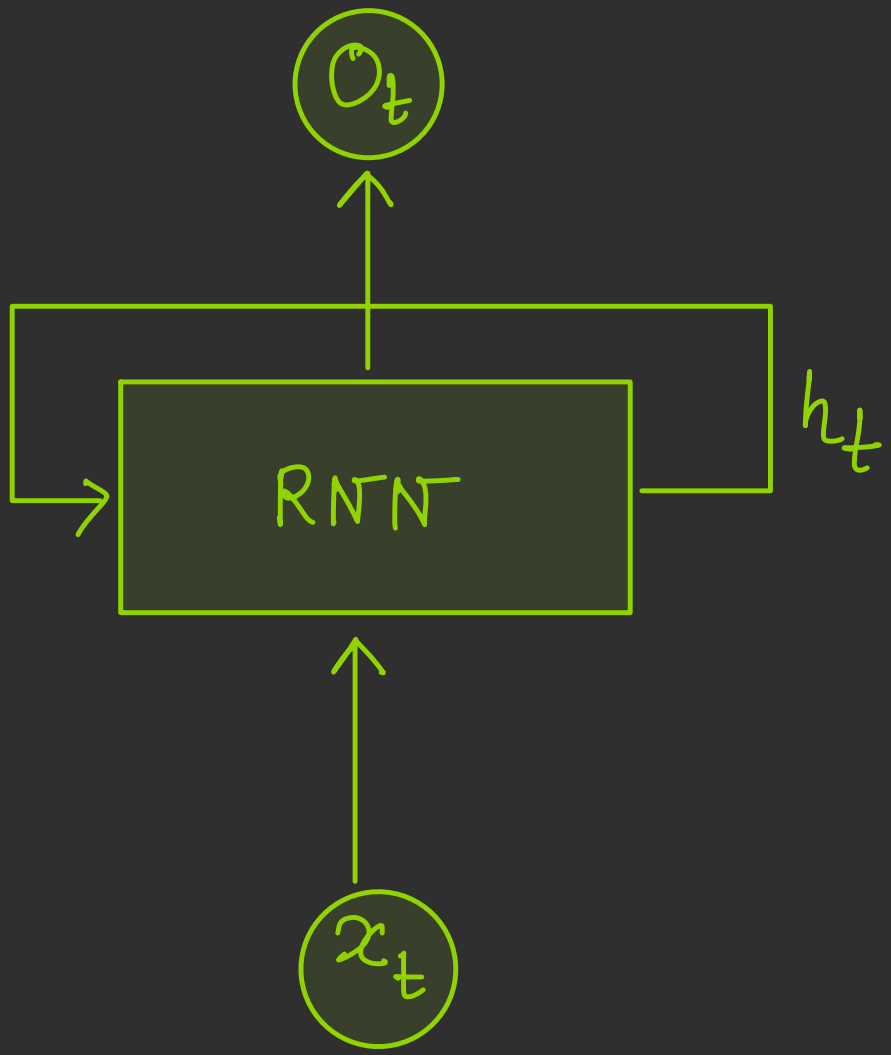
King need money now

Money is shorable  Money is shable [0 0 0 0]

→ padding (zero padding)


[1000]
[0101]
[0101]
[0001]


Hello may
Hello, how are you?
ATTN
how you hello are?




$h_t$ = hidden state

$O_t$ = output

$x_t$ = input

$$O_t = f(x_t, h_{t-1})$$

$t$ = time step

How are you

$t=0$ $t=1$ $t=2$